

The Tool for Intergenerational Transmission Manual (TITA Toolkit)

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INTRODUCTION

The assessment of intergenerational language transmission plays a crucial factor in identifying the vitality of a language. Without intergenerational transmission (IGT) it is certain that a language will not be passed down to the next generation. With almost one third of human languages facing imminent extinction (Lee & Van Way, 2016), the identification and the assessment of the vitality of languages has become a major priority for many disciplines, including the fields of linguistics, education, public health, ethnobotany, etc.

Language endangerment researchers assess language vitality using one of several scales (e.g., Fishman 1990's Graded Intergenerational Disruption Scale; Lewis & Simons 2010's Expanded Graded Intergenerational Disruption Scale; the UNESCO Framework, Lee & Van Way's 2016 Language Endangerment Index, amongst others), all of which point to IGT as the pivotal point at which a language switches from being positively to negatively appraised. However, current scales of language vitality pose a major problem in that—despite the singular importance of language transmission—no rigorous and reproducible method exists for assessing IGT. Currently, anecdotal evidence is the standard for assessing how much a language is passed down generationally.

In bilingual and multilingual communities, having a method of assessing the likelihood of IGT for minority languages is key in determining its future path. Moreover, communities themselves often are unaware that their children are not fully acquiring a language. It is all too common that community members feel that their children are acquiring a language, whereas the children are only partially acquiring that language (see, for example, the heritage language literature, e.g., Benmamoun, Montrul & Polinsky, 2013). Thus a tool is needed that will provide researchers and communities with quantifiable statistics on a comparable degree of acquisition for each language within a community. This may be used by researchers and/or community members to mobilize resources and social awareness towards efforts to improve the rate of acquisition amongst the youngest generation of speakers.

The TITA Toolkit fills this gap through the implementation of six measurements. The first three measurements of TITA use linguistic surveys, assessing both an adult language profile and a child language profile for child-caregiver dyads, along with a vocabulary estimate conceptually based on the MacArthur-Bates Communicative Development Inventory (CDI), to establish the language environment in the home. The remaining three measurements evaluate language ability in the youngest generation (children) by testing word comprehension, word production, and morphosyntactic understanding through assessment of a key distinguishing feature in the language.

The following sections provide detailed instructions for each measurement. There are links to templates, scoring sheets, and assessment materials saved in the TITA Google Drive that is open and accessible to everyone. The TITA team is here to help, and can advise in developing a research methodology, choosing a distinguishing feature to test, scoring, analysis, and general trouble shooting questions. Please reach out to titainfo@hawaii.edu if you have any questions

or comments, and a team member will get back to you. If you do implement the tool the TITA team would love to hear about it, and, if possible, add the statistical data to our database, which we are continuously managing and improving to ensure that the TITA is statistically sound and accurate.

LANGUAGE SURVEY DESCRIPTION

GOAL: The survey constructs a picture of a child's language environment, proficiency, dominance, and use. It creates this picture through two measures: 1) A linguistic profile self-assessment from adults in the child's family, and 2) An adult's assessment of the child's linguistic profile. These two surveys together provide a comprehensive look at the home linguistic environment of the child.

TARGET GROUP: This survey is intended for the parents/caregivers of children living in the household. Recording the exact age of the child at time of survey will help determine the role of input and access at various stages of language development.

MEASURES: The language survey consists of two parts:

1. [Household Adult Language Profile \(HALP\)](#)
2. [Child Language Profile \(CLP\)](#)

HALP:

- The HALP is filled out by a parent/primary caregiver of the child to assess the linguistic profile of child and adult as a dyad.
- The HALP can be sent to the parent/caregiver or can be conducted orally by the researcher or trained a community member depending on the conventions of the community and/or convenience for the community and the researcher. Conducting the survey orally may allow for better communication as certain questions may require examples to explain.
- The HALP constructs a picture of an adult's language environment, proficiency, dominance, and use. Assesses the following:
 - a. **Demographics:** What is the language environment surrounding the adult?
 - b. **Language proficiency, dominance:** What languages does the adult speak, how well does the adult speak them, and which languages are strongest for the adult?
 - c. **Language use:** For each language an adult speaks, How much is spoken in which contexts (where, when, and with whom). This includes domains such as in the home, outside the home with friends and family, at the market, at work, and at religious institutions.
- The adult participant should be surveyed on the HALP in at least two languages, the target (endangered, minority, etc.) language of research and a secondary dominant language. If the community is regularly in contact with more languages, they can also be included.

CLP:

- One CLP is filled out by the adult who filled out the HALP survey, for each child participating in the assessment.
- The CLP can also be sent to the parent/caregiver or conducted orally by the researcher or a trained community member.

- The CLP pieces together the child's language experience through the following specific areas:
 - **Demographics:** What is the language environment surrounding the child?
 - **Language onset:** Which languages does the child speak and understand? When did language production first occur for each language spoken by the child?
 - **Language proficiency, dominance:** What languages does the child speak, how well does the child speak them, and which languages are strongest for the child?
 - **Language use:** How much of each language is spoken by the child among adults and children inside and outside the home?
- CLP survey should be carried out for each language spoken by the child-parent dyad, minimally, the target community language of interest and the larger, dominant language which is in competition (e.g. Kosraean & English).

DATA

The questions on the HALP and CLP result in both qualitative and quantitative data. The qualitative data reports the languages spoken and heard by both members of the child-caregiver dyad along with some attitudes held by the parent regarding language use and importance.

The quantitative data records language usage and proficiency by both the adult and child. These components are used to generate an intergenerational transmission (IGT) score (as described below). These components may (though need not) be used in quantitative data analysis such as ANOVA comparisons, linear/ordinal regression, etc.

Further information regarding the amount of exposure to media is also collected which can be used in both qualitative and quantitative ways.

SCORING FOR IGT:

Each language being assessed by the HALP and CLP will get an IGT score, which we can use to compare the IGT levels of each language.

The following components are scored on the HALP:

- Likert Scale questions (1-5):
 - Adult Language Proficiency
 - Amount of Language Spoken at Home
- Domain Comparisons (0-2)
 - Up to 6 questions asking for language usage of the target or dominant language spoken in domains other than the home scored as 2 (more than the home), 1 (same as the home), 0 (less than the home).

The total possible value for the HALP measure is between 2 and 22 if all six variables for domains are used.

The total possible value for the CLP measure is between 8 and 43 if all three variables for domains are used.

INTERPRETATION OF SCORING:

- Higher scores indicate stronger IGT relative to a language with a lower score.
- There are more questions in the CLP than the HALP. However, the scoring system is not based on the overall amount of questions, but on the overall difference in points between languages. Through this assessment, we can compare directly the IGT of a majority language to Language X (and Y, Z, etc.)
- There are more questions in the CLP than the HALP. However, the scoring system is not based on the overall amount of questions, but on the overall difference in points between languages. Through this assessment, we can compare directly the IGT of a majority language to Language X (and Y, Z, etc.)
- The first set of scores we get is the ratio of the adult score from the HALP($x/22$) and a ratio of the child score from the CLP ($y/43$)
 - A higher ratio indicates more familiarity with the language
 - If either the child or the adult has a higher ratio in the dominant language than the target, minority language, that might indicate vulnerability in the minority language
- The second score we can calculate is a ratio of the child knowledge by the adult knowledge [$(y/43)/(x/22)$]
 - A higher ratio is more intergenerational transmission
 - If this ratio is higher for the dominant language than the community language, the minority language might be vulnerable in terms of transmission

Utilizing the [TITA Survey Data Template](#) and the accompanying [R script](#) will help organize and analyze the data.

Surveys

[HALP](#)

[CLP](#)

Resources to Analyze Data

[TITASurveyData_Template](#)

[R script](#)

CHILD VOCABULARY ASSESSMENT (CVA)

GOAL: The CVA has as its goal the assessment of a child's ability to produce certain vocabulary, as reported by a caregiver who knows the child very well. The CVA is based upon decades of research with similar tools, and is considered a reliable method to estimate a child's vocabulary. It is simply a list of words (constructed as outlined below) which the primary caregiver of the child responds to, indicating whether the child can understand/speak each word.

TARGET GROUP: The CVA is intended to assess vocabulary of children between 1-6 years of age. It is administered to the parent or primary caregiver (e.g., a grandparent or aunt/uncle) who is the person who spends the most time with the child. This CVA is optimized for younger children but may also include older children, though the latter will be less informative. We recommend that the CVA be prioritized for younger children, but that it remain optional, depending on the capacity of the researcher.

OVERVIEW OF CVA:

- Consists of 200 words
- The 200 words are broken up into six semantic categories between 25 and 50 words in each category.
- Words are roughly divided into higher frequency words and lower frequency words.
- Frequency of the words is estimated from native speaker consultants, if no corpus is available.

HOW TO CONSTRUCT THE CVA: A list of 200 words must be constructed, typically with the help of one or more native speaker consultants. The words should fall into the following semantic categories, in the amounts indicated in parentheses.

- 1) Nature and Animals (50 words), broken roughly into the following three subcategories.

Domesticated Animals (10 items): 10 salient domesticated animals. If the community does not make use of domesticated animals, or there are fewer than 10 types of domesticated animals present, additional wild animals may be substituted. For all animal terminology, if the language has separate lexical items for male and female animals and the generic is rarely used (e.g. "billy" and "nanny" are common words, but "goat" is not), include both the male and female terms, but not the generic.

Wild Animals (15 items): 15 salient wild animals. If there are not enough suitable lexical items to meet this guideline, any 15 wild animals may be used; if there are not 15 wild animals in the community (e.g. in a dense urban center), substitute domesticated animals or, failing that, additional landscape terminology.

General Nature Terms (25 items): 25 words for common plants, landscape features, and natural phenomena which will be familiar to children in the

community. Do not select terms which are exceedingly rare (e.g. “snow” in an area where snowfall occurs every three or four years). If the community is in a highly urban area where man-made landscape features are more salient than natural ones, landscape feature terms may be replaced by terms in a similar semantic domain (e.g. substituting “freeway” for “river”).

- 2) Food and Drink (25 words), comprised of words for items such as starch staples, animal products and foraged/cultivated food plants;
- 3) Household (50 words), This section includes common objects for eating, cooking, and food preparation implements; furniture; clothing; toys; tools and vehicles; cleaning supplies; cultural and arts items; and parts of the dwelling;
- 4) People and Body Parts (25 words) includes terms for immediate family members such as mother and father, words for people in general such as man or woman, and common body parts;
- 5) Descriptive Words (25 words), including those words that can be used to describe the world such as very basic color terminology and states of being;
- 6) Action Words (25 words), composed of words relating to concepts of movement or action, such as is found in common parental commands.

FREQUENCY OF WORDS: The words in each category should be divided approximately into 70% higher frequency words and 30% lower frequency words. The best way to do this is to analyze a corpus, ideally a corpus of child directed speech, or of adult-to-adult speech. If not available, a corpus of the written variety of the language may be used (e.g., newspapers). And if no corpus is available, a native speaker consultant should assist. The consultant can be asked to categorize words as either higher frequency (common words) or lower frequency (less common words). For example, if done in English, a consultant would likely categorize the word for *dog* as higher frequency, but the word for *goat* as lower frequency.

Another way to estimate frequency is to think of higher frequency words as those that a 4-5 year old child would likely know, and lower frequency words as those that a 4-5 year old child is unlikely to know, but that a 9-10 year old child is likely to know. This may help the consultant conceive of how to respond to questions about frequency.

It is recommended to follow the [CVA Template](#) as a customizable guide to ensure the categories and frequency ratios required are adhered to.

HOW TO ADMINISTER

- May be administered in person or asynchronously (e.g., Zoom).
- Remember this assessment is filled out by the adult and not the child, so the child should not be consulted on the CVA.

- The researcher may give the care-giver the CVA list and ask them to fill it out themselves.
- If done asynchronously, the researcher should provide the CVA in a format accessible to the care-giver (paper, electronic), and then show them how to fill it out.
- All words should be responded to, and if there is doubt about whether the child can understand or say a word, then it should be responded to in the negative.

SCORING THE CVA: Each word which the child is reported to be able to say is worth one point, for a maximum of 200 possible points. The [CVA Scoring Template](#) can be used to organize and analyze the data.

INTERPRETATION OF THE RESULTS:

- A score of 190-200 indicates strong foundation
- A score of 160-190 indicates medium foundation
- A score of 100-160 indicates low foundation
- A score of less than 100 indicates vulnerability

These numbers may seem high, but a list of 200 basic vocabulary items should be mastered fairly well in order to count as high mastery.

Scores of high-frequency and low-frequency items (weighted the same in scoring) can be recorded separately. For children that score below 160, the researcher may analyze whether the high frequency or low frequency words are more vulnerable. If the child shows weakness in high frequency words, this is more concerning than if the child shows weakness in low frequency words. Moreover, strength in low frequency words is an indicator of vitality.

Resource to Prepare and Implement Tasks

[CVA Template](#)

Resource to Analyze Data

[CVA Scoring Template](#)

WORD COMPREHENSION & WORD PRODUCTION TASKS

GOAL: The lexical tasks are direct measures of a child's basic vocabulary through passive and active knowledge. It tests the speaker's lexicon using a word comprehension task and a word production task with visual and linguistic stimuli. These tasks together provide a broad view of a child's vocabulary and also serve to inform the results of the survey tasks.

TARGET GROUP: The word comprehension task and word production task are aimed at children between 2-10 years of age, though older participants may also be included. The exact range is dependent on the language environment of the children in your specific community. We recommend that a control group of 5-10 adults also be included as a baseline for strong control of vocabulary. If the adult control group scores poorly on these tasks, we may be past the initial stages of language loss.

MEASURES: The word tasks consists of two parts:

1. Word comprehension task
2. Word production task

MATERIALS:

1. [Set A Pictures](#) (Word comprehension task)
2. [Set B Pictures](#) (Word production task)
3. [Word List Scoring Sheet](#)
4. Audio and/or video recorder (optional)

ROLES:

1. Child Participant
2. Interactor (Researcher 1)
3. Recorder (Researcher 2)

It is recommended to have two researchers involved in administering the tasks—one as the interactor and one as the recorder—and at least the Interactor needs to be proficient in the minority language, though both roles can be completed by one researcher.

If available, recording the tasks on an audio-recorder is recommended to facilitate double checking for accuracy and acceptability of the results.

1. CREATION OF A WORD LIST: Both tasks in this section are based on the use of pictures to elicit responses from the child participant. A total of 100 pictures will be used in these tasks. These 100 pictures are divided into two sets of 50 pictures, labelled Set A for the comprehension task and Set B for the production task, each set tests the same 50 word items but with different pictures to prevent priming effects.

The items to be represented in each picture were chosen from the TITA CVA list, see Table 1. To test a wide variety of semantic domains, the items are proportionally distributed across the categories and subcategories of the TITA CVA, see column 5 of Table 1.

Table 1: Example items for each picture task

#	Item	#	Item	Category
1	chicken	2	pig	nouns, nature and animals, domestic animals
3	goat	4	dog	nouns, nature and animals, domestic animals
5	rat	6	ant	nouns, nature and animals, wild animals
7	fish	8	snake	nouns, nature and animals, wild animals
9	sun/moon	10	rain	nouns, nature and animals, nature
11	bone	12	lightning	nouns, nature and animals, nature
13	soil	14	river/ocean	nouns, nature and animals, nature
15	knife	16	fire	nouns, household, food preparation
17	sarong/pants (typical garment)	18	hat/scarf/headwear	nouns, household, clothing
19	broom	20	trash/garbage	nouns, household, cleaning supplies
21	ball/toy	22	doll	nouns, household, toys
23	bed	24	cultural item	nouns, household, furniture & cultural and religious items
25	car/canoe/sled (common transport)	26	fence	nouns, household, tools and vehicles
27	milk	28	egg	nouns, food, drinks & raw foods/ingredients
29	rice/corn/manioc (starch staple crop)	30	breadfruit/onion (common vegetable or fruit)	nouns, food, raw foods/ingredients
31	prepared dish	32	bread/porridge (prepared staple starch)	nouns, food, prepared foods
33	grandparent/elder (most common term of address)	34	child	nouns, people and body parts, people
35	arm	36	fingernail	nouns, people and body parts, body parts
37	eye	38	hair	nouns, people and body parts, body parts
39	cry	40	sleep	action words
41	eat	42	wash	action words

43	break	44	hide	action words
45	big	46	old	descriptive words
47	wet	48	empty	descriptive words
49	sad	50	tired	descriptive words

To create your word list it is recommended to utilize the item list in Table 1 as a foundation and customize it in two ways for the target minority language. One, replace items to suit the community and language being assessed, preferably from the same category as the original. For example, the item 'milk' (#27) can be replaced with 'banana' to maintain cultural and environmental relevance. Two, the vocabulary should be selected to retain a 70:30 ratio of high-frequency and low-frequency words for each category. For example, the nature and animals category includes 14 items so aim to include ten high frequency and four low frequency items. The high frequency words are vocabulary that younger children should have mastered due to their common usage around the home and daily life. The low-frequency words are lexical items normally acquired by older children. The division of the vocabulary helps to assist in determining if children are acquiring words they would use in an environment where the target language would be spoken on a daily basis in all domains. If a naturalistic corpus has not been created, the approximate frequency can be determined by consulting adult speakers to independently sort the frequency of the words into the high or low category.

Colored pictures are depicted in each of the 50 lexical items, because compared to black and white drawings, illiterate and young populations are prone to produce and comprehend tasks utilizing colored pictures much better (Reis et al., 1994, 2001, 2006) and the color detail enhances object recognition (Naor-Raz et al., 2003, Price and Humphreys, 1989, Tanaka et al., 2001, Wurm et al., 1993). Furthermore, color pictures have been evidenced to be recalled faster than black-and white pictures (Davidoff and Ostergaard, 1988). While not recommended black and white pictures are permissible if colored pictures are not feasible.

A template for the creation of your own word list can be found [here](#). Our established [Word Tasks Picture Repository](#) and [Info Sheet](#) provides resources and additional information to compose the tools for this task.

2. COMPREHENSION TASK

Materials: The materials for this task consist of 50 pictures ([Set A](#)). It is recommended to use the [WordList ScoringSheet](#) as a paper or digital method to record the responses of the participants. Each participant should have their own score sheet.

Procedure: The Interactor presents two pictures from the same lexical category simultaneously to the participant and provides a linguistic stimulus in the language that is being tested. They ask the participant to point to the picture that shows what they hear. Examples are shown below.

Table 2: Sample linguistic stimuli for the comprehension task

Types of target word	Linguistic stimuli
Nouns	"Which one is X?" "Point to X."
Action words	"Which person is (eating)?"
Descriptive words	"Which one is (wet)?"

For example, the target picture in one item is of a chicken, while the competitor picture is of a pig. The participant is asked, "Which one is the chicken? Point to the chicken." The Recorder then marks on the scoring sheet if the participant selected the target picture or not. The objective is for the participant to choose the picture that corresponds to what is being heard.

The order of the picture-pairs are always shown in the same pairing and order for each participant. For example, items 1 and 2 from Table 1 are presented together, items 3 and 4 together, etc. The participant sees all 50 pictures, but are actively tested on the comprehension of 25 in one session since they are being shown in pairs. It is recommended that each participant be tested twice with this task in each language, once on the comprehension of half the words, and another session on the comprehension of the other half of the words, so the comprehension of all 50 items are tested. Participants are tested in both the minority language and once in the majority language so each participant will see Set A pictures a total of four times, preferably spaced out over several days. If constraints prevent testing all 50 words in both languages (e.g. two sessions per language) then a researcher may test one set of 25 words in the minority language, and the other set of 25 words in the majority language.

Scoring: Each correct identification via a physical response by pointing to the picture of the word the Interactor says is given one point. If a participant chooses the incorrect word or does not respond to a question, no point is given. There is no ambiguity in the scoring of this task.

Analysis: There are several ways to analyze this task. For instance, if analyzing each participant, a percentage of their accuracy of the overall task, or a breakdown between the high frequency words and the low frequency words in each language can be analyzed. The percentages can easily be calculated to allow for comparing and contrasting individual scores by language. Additional analyses can be conducted based on the goals of your research.

A simple option to compare the overall acquisition of the minority language to the majority language is to transfer the data from the comprehension word task to the [WordComprehension_ScoringTemplate](#). Utilizing this sheet and the accompanying [R script](#) will help organize and analyze the data to calculate the accuracy of the entire word comprehension task, accuracy of the low-frequency items, and accuracy of the high-frequency items by language.

3. PRODUCTION TASK

Materials: The materials for this task consists of 50 pictures ([Set B](#)). A different set of pictures from the comprehension task is utilized so that the participants are not primed by one set prior to being tested on the other lexical task. It is recommended to use the [WordTasks ScoringSheet](#) as a paper or digital method to input the responses of the participants. Each participant should have their own score sheet.

Procedure: The Interactor presents one picture at a time to the participant, and follows with linguistic stimuli asking participants to name the object or describe the picture, examples are in Table 3. Encouragement of the participant to provide an answer is expected with child participants, but after a long pause, 3–4 seconds, with no response then the Interactor moves on to the next picture stimulus. The Recorder marks on the scoring sheet if the participant responds with the target answer, something else, or no response. Ideally the order of the pictures shown would be randomized for each participant. Participants are tested once in the minority language and once in the majority language, so they will see the Set B pictures a total of two times. A comprehension task and production task of the same language can be conducted in the same session, but there should be several days between each session of different languages.

Scoring: Each correct response is given one point. If a participant provides a response which differs from the target answer, then the researcher(s) must decide if the answer is acceptable. If the participant responds with an unacceptable answer or does not respond to a question, no point is given. Because of the open nature of this task, a different coding and scoring system may need to be defined in order to analyze the results.

Table 3: Sample linguistic stimuli for the production task

Types of target word	Linguistic stimuli
Nouns	“What do you call this?” “What is this?”
Action words	“What is this person doing?”
Descriptive words	“How do you describe this?” “What state is this in?”

Analysis: There are several ways to analyze this task. For instance, if analyzing each participant, a percentage of their accuracy of the overall task, or a breakdown between the high frequency words and the low frequency words in each language can be analyzed. The percentages can easily be calculated to allow for comparing and contrasting individual scores by language. Additional analyses can be conducted based on the goals of your research.

A simple option to compare the overall acquisition of the minority language to the majority language is to transfer the data from the comprehension word task to the [WordComprehension_ScoringTemplate](#). Utilizing this sheet and the accompanying [R script](#) will help organize and analyze the data to calculate the accuracy of the entire word comprehension task, accuracy of the low-frequency items, and accuracy of the high-frequency items by language.

Resources to Prepare and Implement Tasks

[WordList_ScoringSheet](#)

[Word Tasks Picture Repository Info Sheet](#)

[Word Tasks Picture Repository](#)

Resources to Analyze Data

[WordComprehension_ScoringTemplate](#)

[WordProduction_ScoringTemplate](#)

[R Script](#)

DISTINGUISHING FEATURE TASK

GOAL: TITA is concerned with detecting the earliest indicators of language loss. Under the (prominent) assumption that language loss occurs first to the features that are unusual for the languages of the world (typologically marked features), the Distinguishing Feature Task provides for a mechanism to capture this initial process of language erosion. **This measure is optional for the administration of TITA**, but we believe it can be very informative as to the strength of a language facing pressure from a more dominant language. The TITA team is available for consultation on the design, implementation and analysis of this task.

OVERVIEW OF DISTINGUISHING FEATURE TEST:

- It may target comprehension or production or both
- It is tailored to the particular feature in question
- There is no required format – the test can be designed by the researcher as appropriate
- The intent is to measure control of a feature which is unique to the target language
- Ideally this feature is (i) distinct from anything in the majority language and (ii) typologically rare.

The task will likely focus on something within the area of syntax (grammar) or phonology (the sound system). The goal is to identify which features are most susceptible to loss and to design a task that can measure whether that feature is being transmitted from the adult generation to the child generation.

The specific goals are to:

- (i) identify the potentially vulnerable feature to be assessed
- (ii) test children on their knowledge of this feature
- (iii) test native speakers on their knowledge of this feature
- (iv) compare (ii) and (iii) as a measure of language loss.

IDENTIFYING TARGET FEATURES

The first step is to consider which features are typologically rare across languages of the world. This may be difficult to determine, since a minority language may have numerous features that are typologically rare. But a short list is all that is needed. Alternatively, a native speaker could be asked which aspects of the minority language children often struggle with. This could yield some potential features for testing.

The second step is to consider the age group you will be targeting. If you are targeting children younger than age 3yrs, then high-level grammatical features are probably not appropriate, but sound- or word- or morphology-related phenomena are more appropriate. See below for more discussion on this point.

But rare phenomena may not be the most appropriate feature to test. For example, if the majority language has a basic word order of sentences that is different from the target language, then perhaps just basic word order of sentences involving subjects, verbs and objects is an appropriate feature to test. For example, Nuosu Yi is a language spoken in China, and has

Subject-Object-Verb word order, while Mandarin has Subject-Verb-Object word order. In this case, a simple elicitation task which requires children to produce transitive sentences might reveal something about the acquisition of Nuosu Yi. Or if the target language has lots of prefixes/suffixes/etc. while the majority language has relatively few, then control over these affixes might be an appropriate feature to test.

The basic idea is to identify some difference between the minority language and the majority language which might be easily testable and which may offer a clue as to whether children are gaining control over that language, independent of the majority language. See the [Distinguishing Features Examples](#) for detailed ideas.

TARGET GROUPS

Ages: We expect that the target group will be children aged younger than 18 years. We also highly recommend, as discussed above, that a group of adult control participants be tested (5-10 adults), as a baseline for how proficient they are in this particular feature. After all, if the adults do not have control over the feature, then we are no longer in the very earliest stages of language loss.

If the target feature is sound related, we recommend recruiting children aged 1yr-4yrs. If the target feature is syntax (grammar) related, we recommend recruiting children aged 5yrs-12yrs. Keep in mind that for syntax-related phenomena, monolingual children acquiring their language under ideal conditions often have not achieved full control of syntax until around age 5-6yrs, so testing younger than that with TITA is likely to provide a misleading picture. Testing children across that full age range will provide the best measure of development in the minority language. These age recommendations are rough guidelines, and older/younger children may be tested, as need be.

Number of participants: This depends on the question being asked, the age span in the study, and whether access to children and adults is easy. First determine the age range in the study, and then break this down into developmental categories. The typical categories used in the field of linguistics for language development are:

	<u>Sound Phenomena</u>	<u>Grammar Phenomena</u>
<1yrs	Perception abilities	n/a
1yrs-2yrs	Production abilities	single words
2yrs-4yrs	Sound combinations	short sentences, often missing grammatical markings (e.g., agreement, tense/aspect)
4-6yrs	Reaching adult-abilities	Development of complex syntax
6-8yrs	Adult-like	Development of pragmatics/discourse
8-10yrs	Adult-like	Adult-like

If the age range being tested spans across multiple age categories, we recommend trying to include at least ten children per age category. For example, if the age range being tested is 4-10yrs, we recommend 10 children aged 4-6yrs, 10 children aged 6-8yrs, and 10 children aged 8-10yrs. If children over age 10yrs are being tested, an additional 10 children of that age group should be tested, unless there is reason to test more children.

Note these are recommendations, but how many children are recruited is often determined by communal and practical concerns. A smaller test population does not invalidate the use of TITA.

Other considerations: One major concern is the onset of formal schooling, often done in the majority language. We recommend trying to straddle the age at which children begin formal schooling, to help assess whether formal schooling is affecting the transmission of the minority language. See, for example, Nusartlert (2023) who found that Isaan children showed a dip in accuracy in lexical comprehension and production at ages 6-8 years old - a fact Nusartlert attributed to the onset of formal schooling in the dominant Thai language.

Moderating the effect of external factors: There may be other factors within the community which affect your selection of participants. For example, if the community in question is a diasporic one, older children might be more appropriate. If the community is a very stable one (linguistically), younger children might be appropriate. Children older than 12 years old may also be tested, as well as older adults, as a comparison to the younger children being discussed here. TITA permits changes of this sort; feel free to contact the TITA team if you have questions.

TESTING CONSIDERATIONS: Consider how long you can test young children for. Under the best conditions, children younger than age 6yrs should not be tested for more than 20 minutes at a time. Children aged 6yrs-10yrs can be tested for slightly longer, and adults can (probably) be tested for up to 40 minutes. This will constrain how long your test is. If you design a simple picture selection task (see below), we recommend having at least 20 items in total, broken down into the appropriate conditions. For example, if testing if children understand singular-plural morphology, we might have 10 items testing knowledge of singular and 10 items testing knowledge of plural. This will keep you well under the 20 minute limit of children's attention. Ideally, the test instrument used to assess children will be the same (or similar) to that used for older children and adults. This will allow for comparison between adults and children.

MEASURES: The measures will greatly depend on which features are being tested and what the methodology is. Below we have some examples of features that may be tested and the methodologies used to test, though TITA permits flexibility in this regard.

ANALYSIS: The analysis of the results of this task will depend on the design of the task and the feature being tested. Ideally, the task will be designed in a simple manner, either comprehension or production,, but designed to gather simple data which can be analyzed without the need for sophisticated analysis methods or tools. We recommend picture selection and picture

description as simple methods that do not demand high expertise in experimental design. But if more ambitious methods are needed, the TITA team is available for consultation.

Table 4 provides some examples of distinguishing features in specific language scenarios which could prove fruitful for inclusion in TITA.

Table 4: Overview of Distinguishing Feature Examples of various languages

Distinguishing Feature	Language 1	Language 2	Method	Measure
Passives in Thai/Isaan	Isaan: Main Verb Passive: acceptable By-Phrase Passive: ungrammatical	Thai: Main Verb Passive: acceptable By-Phrase Passive: acceptable	Picture Selection Task: Present sentence with two pictures, one matching the passive sentence, the other matching the active version of the sentence.	Accuracy of selecting the matching picture
Implosives	Pokot: explosive and implosive sounds	English: only explosive sounds	Picture Naming Task: picture shown of object whose label includes an implosive sound, child must name the object	Accuracy of production of implosive sound
Modifier Order	Kosraean: post-nominal modifiers , e.g., (the equivalent of) <i>fan one</i>	English: prenominal modifiers, e.g., <i>one fan</i>	Two-picture naming task: two pictures depicting the same object(s), differing in some modifier (e.g., number, color, etc.). One is highlighted and child must name the picture	Accuracy of use of modifier relative to

Inclusive/Exclusive Pronouns	Samoan Inclusive and Exclusive pronouns: 1p Sg a'u 1p Pl In tātou 1p Pl Ex. mātou	English: 1p Sg 'I' 1p Pl 'We'	Truth Value Judgment Task "Va'ai mai, 'olo'ota'a'alo mātou" "Look over there, we're (exclusive) playing"	Rate of accurate rejection of false items.
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FREQUENTLY ASKED QUESTIONS

1. How many children must one test?

The answer to this depends on the size and complexity of the language situation. You want to gather a reasonable amount of data to be able to develop reasonable inferences. A base number to consider would be approximately 10 children per age range that you plan to test (e.g. 2-3 years old, 4-5 years old, 6-7 years old, 8-11 years old). These aren't hard numbers and ultimately the more children that can be assessed across ages (i.e. not all the same age) will always provide a better evaluation of your community's language situation.

2. How are the scores from each measurement aggregated?

There are several scores that can be calculated from each instrument in the TITA toolkit, these subsection scores help identify how the language is being transmitted, what types of vocabulary and features are being transmitted at which age-groups, and which areas might be vulnerable.

A formula is provided in which the scores from each instrument are collected and standardized. Researchers can then report the comparative scores from each language, the target minority language(s) under investigation, and the competing majority language, to assess the relative vitality of the community language.

3. What happens if a researcher is unable to conduct all six measurements?

While the most robust data can be derived from implementing all the measurements, we understand that time and energy does not always allow for that. At minimum, the Adult Language Survey, Child Language Survey, Word Comprehension, and Word Production measurements are required to fully have a comparable view of a language's vitality in a community. The two optional measurements are the Child Vocabulary Assessment and the Distinguishing Feature task.

4. Why are two of the instruments (the CVA and the Distinguishing Feature) listed as optional?

The two surveys (HALP and CLP) are relatively easy to design and administer, as are the two word tasks (word comprehension and word production). These four instruments also provide a comprehensive snapshot of the linguistic environment of the youngest generation, collecting both survey and experimental data. For this reason, the TITA Team recommends making these four instruments the basic foundation of any assessment. The CVA and Distinguishing Feature tasks are more challenging to design and administer, and so we recommend these tasks only if the researcher is able and has the capacity.

CVA is harder to design (requiring a larger frequency analysis of words), and harder to administer (parents are often less enthusiastic to take this survey, as it requires checking 200 words). The Distinguishing Feature task requires some knowledge of linguistics, and may be a challenge to design. Nonetheless, these tasks have the potential to provide invaluable information about IGT, and therefore remain recommended.

Moreover, the CVA is optimized for parents/caregivers who are assessing the vocabulary of children younger than age 6yrs. Beyond that age range, this measure becomes less effective, making it less applicable in all contexts. Similarly, the Distinguishing Feature Task may also be age-restricted. The other tasks within TITA are much more applicable to children of all age ranges.

5. Do word–picture tasks provide an accurate measure of children’s linguistic knowledge?

Studies have shown that picture-based tasks provide an accurate measure of children’s linguistic knowledge across a range of educational backgrounds and developmental stages (Johnson et al., 1996; Reis et al., 2006). This approach is grounded in well-established methods used in first language acquisition research, including standardized assessments such as the Peabody Picture Vocabulary Test (Dunn and Dunn, 2007).

6. Can I implement the TITA remotely?

Yes, the tool can be implemented remotely or in person. While being in person provides an enriched relationship and trust within the community, if that has already been established it is feasible to conduct the tests remotely. Though it would be prudent to have at least one community member who is a native speaker in the language assist the children being tested when implementing any comprehension and production tools. When conducting the research the most important aspects are to keep the data organized and test children in a quiet place where disruptions and distractions are kept to a minimum, whether in person or remotely.

7. Can a non-native speaker of a language implement TITA?

Yes, but it is recommended to involve members of the community as consultants, not only in developing your word lists and instruments, but also to assist in implementing the tools. At least one researcher should be a native speaker of the language(s) to test children in the minority language(s) being assessed.

8. Has the TITA toolkit been validated?

The TITA is currently undergoing validation and we have conducted test validation assessments for both of the surveys to ensure they work well in different communities. We are working on validating the word tasks with continuing data. As both the CVA and the Distinguishing Feature instruments are optional and can be highly specialized, validation of those tasks is not currently being conducted.

9. What if I don’t know how to use R to analyze the data?

You do not need R to analyze the data, we provide detailed information about how to score and interpret the assessments. But if you’ve never used R before and want to try it for your analysis and don’t know where to start please get in touch with the TITA team! We understand it can be intimidating at first. There are demo spreadsheets as test examples in the R folder of the toolkit. We would be happy to help go through it and/or show how to edit the script for the analysis you want.

REFERENCES

- Benmamoun, E., Montrul, S. and Polinsky, M. (2013a), 'Keynote article. Heritage languages and their speakers: opportunities and challenges for linguistics', *Theoretical Linguistics*, 39, 129–181.
- Brezinger, M., A. Yamamoto, N. Aikawa, D. Koundioubu, A. Minasyan, A. Dwyer, C. Grinevald, M. Krauss, O. Miyaoka, O. Sakiyama, R. Smeets, & O. Zepeda. 2003. *Language vitality and endangerment*, Paris, UNESCO Ad Hoc Expert Group Meeting on Endangered Languages. <https://ich.unesco.org/doc/src/00120-EN.pdf>
- Davidoff, J. B., & Ostergaard, A. L. 1988. The role of colour in categorical judgements. *The Quarterly Journal of Experimental Psychology* 40(3), 533–544.
- Dunn, L.M. & D.M. Dunn. 2007. *Peabody picture vocabulary test* (4th ed.). Circle Pines: American Guidance Service.
- Fishman, Joshua A. 1991. *Reversing language shift: Theoretical and empirical foundations of assistance to threatened languages*. Vol. 76. Clevedon: Multilingual Matters.
- Johnson, C. J., A. Paivio, & J.M. Clark. 1996. Cognitive components of picture naming. *Psychological Bulletin* 120(1). 113–139. <https://doi.org/10.1037/0033-2909.120.1.113>
- Lee, Nala Huiying & John Van Way. 2016. Assessing levels of endangerment in the Catalogue of Endangered Languages (ELCat) using the Language Endangerment Index (LEI). *Languages in Society* 45(2). 271–292.
- Lewis, M. Paul & Gary F. Simons. 2010. Assessing endangerment: Expanding Fishman's GIDS. *Revue Roumaine de Linguistique* 55(2). 103–120.
- Muagututi'a, Grant T. 2018. Recovering Ergativity in Heritage Samoan. [Doctoral dissertation, University of Hawai'i at Mānoa]. ProQuest Dissertations & Theses Global.
- Nusartlert, Anongnard. 2023. Isan-Thai Diglossia: Intergenerational Transmission as the Key Indicator to Vitality. [Doctoral dissertation, University of Hawai'i at Mānoa]. ProQuest Dissertations & Theses Global.
- Naor-Raz, G., M.J. Tarr & D. Kersten. 2003. Is color an intrinsic property of object representation? *Perception* 32(6). 667–680. <https://doi.org/10.1068/p5050>
- Price, C. J. & G.W. Humphreys. 1989. The effects of surface detail on object categorization and naming. *Quarterly Journal of Experimental Psychology* 41(4). 797–828.
- Reis, A., L. Faisca, M. Ingvar & K.M. Petersson. 2006. Color makes a difference: two-dimensional object naming in literate and illiterate subjects. *Brain and Cognition* 60(1). 49–54. <https://doi.org/10.1016/j.bandc.2005.09.012>
- Reis, A., M. Guerreiro & A. Castro-Caldas. 1994. Influence of educational level of non brain-damaged subjects on visual naming capacities. *Journal of Clinical and Experimental Neuropsychology* 16(6). 939–942. <https://doi.org/10.1080/01688639408402705>
- Reis, A., K.M. Petersson, A. Castro-Caldas & M. Ingvar. 2001. Formal schooling influences two- but not three-dimensional naming skills. *Brain and Cognition* 47(3). 397–411. <https://doi.org/10.1006/brcg.2001.1316>
- Sou, Jennifer. 2022. The Acquisition of Kosraean and English: A Diglossic Community at a Crossroads. [Doctoral dissertation, University of Hawai'i at Mānoa]. ProQuest Dissertations & Theses Global.

- Tanaka, J., D. Weiskopf & P. Williams. 2001. The role of color in high-level vision. *Trends in Cognitive Sciences* 5(5). 211–215. 10.1016/s1364-6613(00)01626-0
- Wurm, L. H., G.E. Legge, L.M. Isenberg & A. Luebker. 1993. Color improves object recognition in normal and low vision. *Journal of Experimental Psychology: Human Perception and Performance* 19(4). 899–911.<https://doi.org/10.1037/0096-1523.19.4.899>